Q3. Consider the E-R diagram of Bank Database. Each bank can have multiple branches and each branch can have multiple account and loans. Code, Acct-no, P_no, and loan_no are underlined as primary keys. **[10 Marks]**

a) List strong entity types in the E-R diagram

b) Is there any weak entity? If so give it name, identify distriminator name and name of the identifying relationship.

c) List name of all relationship types.

d) Find the minimum number of tables required to represent the following ER diagram in a relational model. Also identify what will be attributes of these tables

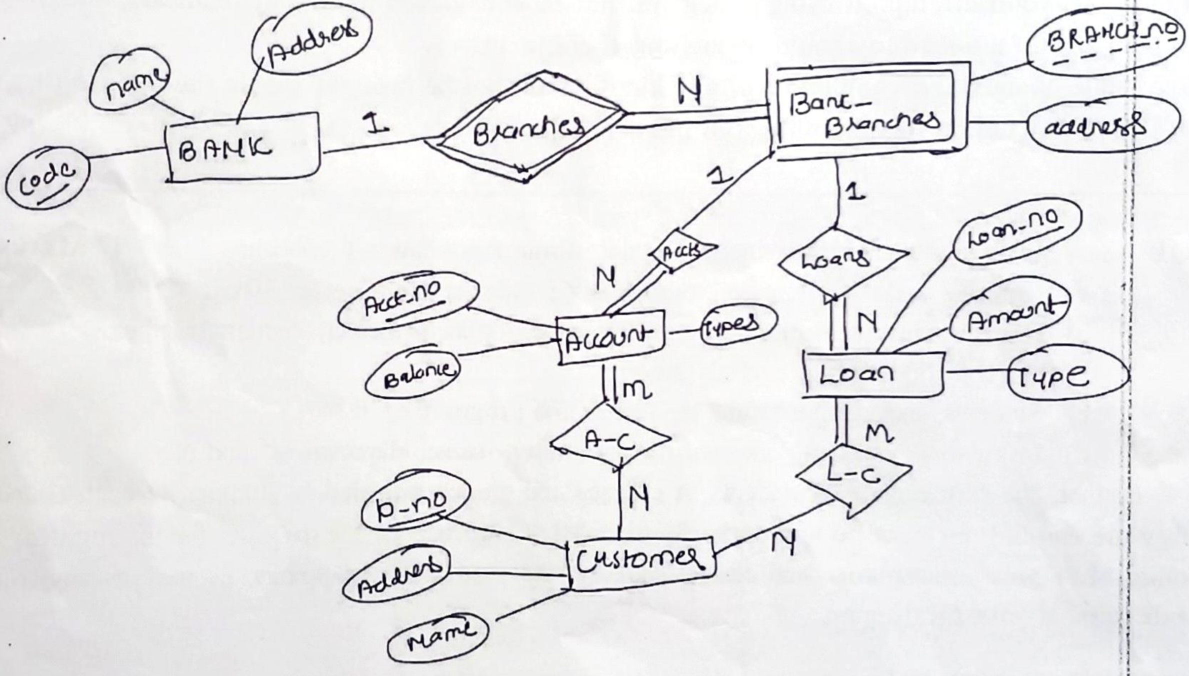

Fig.1. Bank Database

Q4. a) What is the different level of data abstraction? Explain each of them with proper diagram?

b) What is the difference between stored and derived attributes? **[3*2=6 Marks]**

c) Define the concept of aggregation with the help of an example.

Mahindra University Hyderabad

École Centrale School of Engineering

Minor-I exam

Program: B. Tech. **Branch: ECM, ICS, CSE, AI, CM, COM** **Year: 3rd** **Semester: I**

Subject: Database Management (CS 3103)

Date: 11-09-2025 **Start Time: 2:00 PM**

Time Duration: 1.5 Hours **Max. Marks: 30 Marks**

Instructions:

1) Any question attempted using pencil will not be considered for the evaluation

2) All parts of a question should be answered consecutively.

3) Mobile phones and computers of any kind should not be brought inside the exam hall.

4) Use of any unfair means will result in severe disciplinary action.

Q1. A university registrar's office maintains data about the following entities: **[7 Marks]**

a) Courses, including number, title, credits, syllabus, and prerequisities;

b) Course offerings, including course_number, year, semester, section_number, instructor(s), timing, and classroom;

c) Students, including student_id, name, and program;

d) Instructors, including identification_number, name, department, and title.

Further, the enrollement of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an ER diagram for the registrar's office. List your assumptions and clearly indicate the cardinality mappings as well as any role indicators in your ER diagram.

Q2. Consider the following case study and design the EER diagram using the concept of generalization and specialization indicating the overlapping and disjoint constraints along with the total and partial participation. Following are the details of the superclass and subclass along with the attributes. **[7 Marks]**

A university maintains information of each Person having Name (consisting of First_name, Middle_name, Last_name), Address (consisting of Street, City, Pincode), Gender, DoB, and a unique Person_id. Person are further categories as Employee, Alumnus, Student. For each employee the following information is maintained, i.e., Date_hired, salary. For each Alumuns the following information is maintained, i.e., Degree which can hold two values i.e., Year and Date. Each employee is further divided into two categories, i.e., Faculty and Staff. For each Faculty we maintain the information in terms of Rank and for each Staff we maintain the information in terms of their Position. Likewise, each Student is further divided into two categories, i.e., Undergraduate_Student, Postgraduate_Student. For each Postgraduate_Student we maintain the information in terms of Test_score and for each Undergraduate_Student we maintain the information in terms of Class_standing_performance.